



Determine what data to collect

Welcome back. We've talked a lot about all the data out there in the world. But as a data analyst, you'll need to decide what kind of data to collect and use for every project. With a nearly endless amount of data out there, this can be quite a bit of a data **dilemma**, but there's good news. **In this video, you'll learn which factors to consider when collecting data.** Usually, you'll have a head start in figuring out the right data for the job, because the data you need will be given to you, or your business task or problem will narrow down your choices. Let's start with a question like, what's causing increased rush hour traffic in your city? **First, you need to know how the data will be collected.** You might use observations of traffic patterns to count the number of cars on city streets during particular times. You notice that cars are getting **backed up** on a specific street. That brings us to data sources. In our traffic example, your observations would be **first-party data**. **This is data collected by an individual or group using their own resources.** Collecting first-party data is typically the preferred method because you know exactly where it came from. You might also have **second-party data, which is data collected by a group directly from its audience and then sold.** In our example, if you aren't able to collect your own data, you might buy it from an organization that's led traffic pattern studies in your city. This data didn't start with you, but it's **still reliable** because it came from a source that has experience with traffic analysis. The same can't always be said

about third-party data or data collected from outside sources who did not collect it directly. This data might have come from a number of different sources before you investigated it. It might not be as reliable, but that doesn't mean it can't be useful. You'll just want to make sure you check it for accuracy, bias, and credibility. Actually, no matter what kind of data you use, it needs to be inspected for accuracy and trustworthiness. We'll learn more about that process later. For now, just remember that the data you choose should apply to your needs, and it must be approved for use. As a data analyst, it's your job to decide what data to use, and that means choosing the data that can help you find answers and solve problems and not getting distracted by other data. In our traffic example, financial data probably wouldn't be that helpful, but existing data about high volume traffic times would be. Okay. Now let's talk about how much data to collect. In data analytics, a population refers to all possible data values in a certain data set. If you're analyzing data about car traffic in a city, your population would be all the cars in that area. But collecting data from the entire population can be pretty challenging. That's why a sample can be useful. A sample is a part of a population that is representative of the population. You might collect a data sample about one spot in the city and analyze the traffic there, or you might pull a random sample from all existing data in the population. How you choose your sample will depend on your project. As you collect data, you'll also want to make sure you select the right data type. For traffic data, an appropriate data type could be the dates of traffic records stored in a date format. The dates could help you figure what days of the week there is likely to be a high volume of traffic in the future. We'll explore this topic in more detail soon. Finally, you need to determine the time frame for data collection. In our example, if you needed an answer

immediately, you'd have to use historical data, which is data that already exists. But let's say you needed to track traffic patterns over a long period of time. That might affect the other decisions you make during data collection. Now you know more about the different data collection considerations you'll use as a data analyst. Because of that, you'll be able to find the right data when you start collecting it yourself. There's still more to learn about data collection, so stay tuned.

Data collection considerations

- How the data will be collected
- Choose data sources
- Decide what data to use
- How much data to collect
- Select the right data type
- Determine the time frame